



APPLIED LINGUISTICS GRADUATE PROGRAMME (LAEL)

Images function set - Troubleshooting for Yara

The purpose of this document is to register the findings of the troubleshooting process for Yara. The approach consists of reproducing the outcomes on a separate environment and comparing with the ones obtained by Yara.

Please refer to the following repository:

- [cl_images_yara](#)

Functions 'presample' to 'collecturls'

- The same 'urls.txt' that Yara obtained could be reproduced - they are identical

```
ubuntu@ip-172-31-46-151:~/my_env$ cd cl_images_yara
ubuntu@ip-172-31-46-151:~/my_env/cl_images_yara$ ls -la
total 20504
drwxrwxr-x 3 ubuntu ubuntu 4096 Nov 18 10:13 .
drwxrwxr-x 9 ubuntu ubuntu 4096 Nov 18 09:57 ..
drwxrwxr-x 8 ubuntu ubuntu 4096 Nov 18 10:13 .git
-rw-rw-r-- 1 ubuntu ubuntu 42 Nov 18 09:57 README.md
-rw-rw-r-- 1 ubuntu ubuntu 5390025 Nov 18 10:13 images_index_after_dedupe_yara.txt
-rw-rw-r-- 1 ubuntu ubuntu 5594224 Nov 18 10:13 images_index_before_dedupe.txt
-rw-rw-r-- 1 ubuntu ubuntu 4995295 Nov 18 09:59 urls.txt
-rw-rw-r-- 1 ubuntu ubuntu 4995295 Nov 18 09:57 urls_yara.txt
ubuntu@ip-172-31-46-151:~/my_env/cl_images_yara$
```

- So far, the 'images_index.txt' that was obtained has a number of lines that matches the index: 040065
- It contains more lines than Yara's one, which means that further processing in other functions will remove images that are not suitable such as duplicates and empty files
- 50 folders to store the images were created as expected

```

images_index_after_dedupe_yara.txt  images_index_before_dedupe.txt  images_index_after_dedupe_yara.txt X
images_index_after_dedupe_yara.txt
40036 fl:41|t:040036|id:1628173229188649000|d:2023-02-21|u:pacchinirafae 38576 fl:41|t:040036|id:1628173229188649000|d:2023-02-21|u:pacchinirafae
40037 fl:41|t:040037|id:1628173101002268700|d:2023-02-21|u:pacchinirafae 38577 fl:41|t:040037|id:1628173101002268700|d:2023-02-21|u:pacchinirafae
40038 fl:41|t:040038|id:1628172862937784300|d:2023-02-21|u:baianinhoii|i 38578 fl:41|t:040038|id:1628172862937784300|d:2023-02-21|u:baianinhoii|i
40039 fl:41|t:040039|id:1628172861876699100|d:2023-02-21|u:luizcsfh|i:ht 38579 fl:41|t:040039|id:1628172861876699100|d:2023-02-21|u:luizcsfh|i:ht
40040 fl:41|t:040040|id:1628171880208253000|d:2023-02-21|u:djmarkguedes| 38580 fl:41|t:040040|id:1628171880208253000|d:2023-02-21|u:djmarkguedes|
40041 fl:41|t:040041|id:1628171682048364500|d:2023-02-21|u:phannix|i:htt 38581 fl:41|t:040041|id:1628171682048364500|d:2023-02-21|u:phannix|i:htt
40042 fl:41|t:040042|id:1628171322734813200|d:2023-02-21|u:lantaue|i:htt 38582 fl:41|t:040042|id:1628171322734813200|d:2023-02-21|u:lantaue|i:htt
40043 fl:41|t:040043|id:1628171312228098000|d:2023-02-21|u:olivanogueira 38583 fl:41|t:040043|id:1628171312228098000|d:2023-02-21|u:olivanogueira
40044 fl:41|t:040044|id:1628170212607504400|d:2023-02-21|u:solangeferre 38584 fl:41|t:040044|id:1628170212607504400|d:2023-02-21|u:solangeferre
40045 fl:41|t:040045|id:1628169934156038100|d:2023-02-21|u:ciceroferre 38585 fl:41|t:040045|id:1628169934156038100|d:2023-02-21|u:ciceroferre
40046 fl:41|t:040046|id:1628169451559432200|d:2023-02-21|u:lantaue|i:htt 38586 fl:41|t:040046|id:1628169451559432200|d:2023-02-21|u:lantaue|i:htt
40047 fl:41|t:040047|id:1628169339626045400|d:2023-02-21|u:andrecpalto|i 38587 fl:41|t:040047|id:1628169339626045400|d:2023-02-21|u:andrecpalto|i
40048 fl:41|t:040048|id:1628168502405210000|d:2023-02-21|u:mazeto_miriam 38588 fl:41|t:040048|id:1628168502405210000|d:2023-02-21|u:mazeto_miriam
40049 fl:41|t:040049|id:1628168335979413500|d:2023-02-21|u:bacanaioe|i:h 38589 fl:41|t:040049|id:1628168335979413500|d:2023-02-21|u:bacanaioe|i:h
40050 fl:41|t:040050|id:1628168278601347000|d:2023-02-21|u:claudio755765 38590 fl:41|t:040050|id:1628168278601347000|d:2023-02-21|u:claudio755765
40051 fl:41|t:040051|id:1628168081909448700|d:2023-02-21|u:mazeto_miriam 38591 fl:41|t:040051|id:1628168081909448700|d:2023-02-21|u:mazeto_miriam
40052 fl:41|t:040052|id:162816775565520300|d:2023-02-21|u:roberto0830127 38592 fl:41|t:040052|id:162816775565520300|d:2023-02-21|u:roberto0830127
40053 fl:41|t:040053|id:1628167233305297000|d:2023-02-21|u:malacraiaofic→ 38593 fl:41|t:040053|id:1628167233305297000|d:2023-02-21|u:malacraiaofic→
40054 fl:41|t:040054|id:1628166279277183000|d:2023-02-21|u:damottaff|i:h 38594 fl:41|t:040054|id:1628166279277183000|d:2023-02-21|u:damottaff|i:h
40055 fl:41|t:040055|id:1628166161106931700|d:2023-02-21|u:baptista010do 38595 fl:41|t:040055|id:1628166161106931700|d:2023-02-21|u:baptista010do
40056 fl:41|t:040056|id:1628165352201810000|d:2023-02-21|u:nildomardanta 38596 fl:41|t:040056|id:1628165352201810000|d:2023-02-21|u:nildomardanta
40057 fl:41|t:040057|id:1628165101172756500|d:2023-02-21|u:lantaue|i:htt 38597 fl:41|t:040057|id:1628165101172756500|d:2023-02-21|u:lantaue|i:htt
40058 fl:41|t:040058|id:1628164792044101600|d:2023-02-21|u:semprelolyta| 38598 fl:41|t:040058|id:1628164792044101600|d:2023-02-21|u:semprelolyta|
40059 fl:41|t:040059|id:1628164593770963000|d:2023-02-21|u:marcelofdasil 38599 fl:41|t:040059|id:1628164593770963000|d:2023-02-21|u:marcelofdasil
40060 fl:41|t:040060|id:1628163966005305300|d:2023-02-21|u:ivan2013cam|i 38600 fl:41|t:040060|id:1628163966005305300|d:2023-02-21|u:ivan2013cam|i
40061 fl:41|t:040061|id:1628163856726913000|d:2023-02-21|u:moacirmachado 38601 fl:41|t:040061|id:1628163856726913000|d:2023-02-21|u:moacirmachado
40062 fl:41|t:040062|id:1628163552669212700|d:2023-02-21|u:joncarvalhal| 38602 fl:41|t:040062|id:1628163552669212700|d:2023-02-21|u:joncarvalhal|
40063 fl:41|t:040063|id:1628162984580116500|d:2023-02-21|u:joncarvalhal| 38603 fl:41|t:040063|id:1628162984580116500|d:2023-02-21|u:joncarvalhal|
40064 fl:41|t:040064|id:1628162681403330600|d:2023-02-21|u:olivanogueira→ 38604 fl:41|t:040064|id:1628162681403330600|d:2023-02-21|u:olivanogueira→
40065 fl:41|t:040065|id:1628162155894677500|d:2023-02-21|u:joncarvalhal| 38605 fl:41|t:040065|id:1628162155894677500|d:2023-02-21|u:joncarvalhal|
40066

```

Function 'grabimages'

- Inspecting the images, there are empty files. They are supposed to be detected and removed from 'images_index.txt'

```

ubuntu@ip-172-31-46-151:~/my_env/cl_images/images/images/01$ ls -l
total 112948
drwxrwxr-x  2 ubuntu ubuntu  36864 Nov 18 10:22 .
drwxrwxr-x 52 ubuntu ubuntu   4096 Nov 18 09:03 ..
-rw-rw-r--  1 ubuntu ubuntu  47935 Nov 18 10:18 000001.jpg
-rw-rw-r--  1 ubuntu ubuntu 124709 Nov 18 10:18 000002.jpg
-rw-rw-r--  1 ubuntu ubuntu 108603 Nov 18 10:18 000003.jpg
-rw-rw-r--  1 ubuntu ubuntu  48152 Nov 18 10:18 000004.jpg
-rw-rw-r--  1 ubuntu ubuntu  88757 Nov 18 10:18 000005.jpg
-rw-rw-r--  1 ubuntu ubuntu 173601 Nov 18 10:18 000006.jpg
-rw-rw-r--  1 ubuntu ubuntu 175797 Nov 18 10:18 000007.jpg
-rw-rw-r--  1 ubuntu ubuntu  95849 Nov 18 10:18 000008.jpg
-rw-rw-r--  1 ubuntu ubuntu 168024 Nov 18 10:18 000009.jpg
-rw-rw-r--  1 ubuntu ubuntu  78332 Nov 18 10:18 000010.jpg
-rw-rw-r--  1 ubuntu ubuntu 122467 Nov 18 10:18 000011.jpg
-rw-rw-r--  1 ubuntu ubuntu  94587 Nov 18 10:18 000012.jpg
-rw-rw-r--  1 ubuntu ubuntu  60053 Nov 18 10:18 000013.jpg
-rw-rw-r--  1 ubuntu ubuntu  59327 Nov 18 10:18 000014.jpg
-rw-rw-r--  1 ubuntu ubuntu 204864 Nov 18 10:18 000015.jpg
-rw-rw-r--  1 ubuntu ubuntu 207835 Nov 18 10:18 000016.jpg
-rw-rw-r--  1 ubuntu ubuntu  74123 Nov 18 10:18 000017.jpg
-rw-rw-r--  1 ubuntu ubuntu  56812 Nov 18 10:18 000018.jpg
-rw-rw-r--  1 ubuntu ubuntu  40536 Nov 18 10:18 000019.jpg
-rw-rw-r--  1 ubuntu ubuntu 229946 Nov 18 10:18 000020.jpg
-rw-rw-r--  1 ubuntu ubuntu  28157 Nov 18 10:18 000021.jpg
-rw-rw-r--  1 ubuntu ubuntu 697107 Nov 18 10:18 000022.jpg
-rw-rw-r--  1 ubuntu ubuntu  88402 Nov 18 10:18 000023.jpg
-rw-rw-r--  1 ubuntu ubuntu 428931 Nov 18 10:18 000024.jpg
-rw-rw-r--  1 ubuntu ubuntu 428931 Nov 18 10:18 000025.jpg
-rw-rw-r--  1 ubuntu ubuntu 428931 Nov 18 10:18 000026.jpg
-rw-rw-r--  1 ubuntu ubuntu  42389 Nov 18 10:18 000027.jpg
-rw-rw-r--  1 ubuntu ubuntu      0 Nov 18 10:18 000028.jpg
-rw-rw-r--  1 ubuntu ubuntu  78977 Nov 18 10:18 000029.jpg
-rw-rw-r--  1 ubuntu ubuntu 115866 Nov 18 10:18 000030.jpg
-rw-rw-r--  1 ubuntu ubuntu 542413 Nov 18 10:18 000031.jpg
-rw-rw-r--  1 ubuntu ubuntu 542413 Nov 18 10:18 000032.jpg
-rw-rw-r--  1 ubuntu ubuntu 353998 Nov 18 10:18 000033.jpg
-rw-rw-r--  1 ubuntu ubuntu 120972 Nov 18 10:18 000034.jpg
-rw-rw-r--  1 ubuntu ubuntu  25740 Nov 18 10:18 000035.jpg
-rw-rw-r--  1 ubuntu ubuntu 105210 Nov 18 10:18 000036.jpg
-rw-rw-r--  1 ubuntu ubuntu 105210 Nov 18 10:18 000037.jpg
-rw-rw-r--  1 ubuntu ubuntu  57246 Nov 18 10:18 000038.jpg

```

- While running 'grabimages' the performance of the server degraded gradually to the point of becoming no longer reachable. Eventually, it had to be rebooted. The image collection stopped at image 032258 after a series of empty files which should not have been empty

```

ec2-15-229-49-55.sa-east-1.compute.amazonaws.com ***
-rw-rw-r-- 1 ubuntu ubuntu 172623 Nov 18 21:37 032225.jpg
-rw-rw-r-- 1 ubuntu ubuntu 74073 Nov 18 21:38 032226.jpg
-rw-rw-r-- 1 ubuntu ubuntu 133798 Nov 18 21:39 032227.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 21:39 032228.jpg
-rw-rw-r-- 1 ubuntu ubuntu 61263 Nov 18 21:43 032229.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 21:44 032230.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 21:46 032231.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 21:50 032232.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 21:53 032233.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 21:59 032234.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:04 032235.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:08 032236.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:11 032237.jpg
-rw-rw-r-- 1 ubuntu ubuntu 180814 Nov 18 22:15 032238.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:16 032239.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:19 032240.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:23 032241.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:27 032242.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:31 032243.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:38 032244.png
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:44 032245.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:50 032246.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 22:56 032247.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:02 032248.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:09 032249.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:16 032250.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:22 032251.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:29 032252.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:36 032253.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:46 032254.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 18 23:54 032255.jpg
-rw-rw-r-- 1 ubuntu ubuntu 47936 Nov 19 00:04 032256.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 19 00:04 032257.png
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 19 02:43 032258.jpg
ubuntu@ip-172-31-46-151:~/my_env/cl_images/images/images/33$ █

```

- The image 032258.jpg actually exists and could have been collected
 - fl:33|t:032258|id:1553193928903508000|d:2022-07-30|u:cicero78713658|i:<https://pbs.twimg.com/media/FY4NpX8WAAEh5KM?format=jpg&name=large>|f:jpg



- The image 031983.jpg Yara reported as corrupted could be successfully collected this time



- The evidence shows that 'grabimages' or something in its context is not reliable. The function is going to be re-run in an attempt to obtain better results.

```
-rw-rw-r-- 1 ubuntu ubuntu 173870 Nov 19 16:15 035218.jpg
-rw-rw-r-- 1 ubuntu ubuntu 21044 Nov 19 16:15 035219.jpg
-rw-rw-r-- 1 ubuntu ubuntu 565558 Nov 19 16:15 035220.jpg
-rw-rw-r-- 1 ubuntu ubuntu 259268 Nov 19 16:15 035221.jpg
-rw-rw-r-- 1 ubuntu ubuntu 116790 Nov 19 16:15 035222.jpg
-rw-rw-r-- 1 ubuntu ubuntu 101360 Nov 19 16:15 035223.jpg
-rw-rw-r-- 1 ubuntu ubuntu 46206 Nov 19 16:15 035224.jpg
-rw-rw-r-- 1 ubuntu ubuntu 20900 Nov 19 16:15 035225.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 19 16:15 035226.jpg
-rw-rw-r-- 1 ubuntu ubuntu 0 Nov 19 16:27 035227.jpg
-rw-rw-r-- 1 ubuntu ubuntu 28009 Nov 19 16:52 035228.jpg
ubuntu@ip-172-31-46-151:~/my_env/cl_images/images/images/36$
```

- The server became unreachable again and had to be rebooted again. It has been determined that the issue happened after the collection of image 035225.jpg. Images 035226.jpg and 035227.jpg are not truly empty. This time, the 'grabimages' function will be altered in order to collect images from 035226.jpg on
 - fl:36|t:035226|id:1586858281166856200|d:2022-10-30|u:myaorthammy|i:<https://pbs.twimg.com/media/FgWnMIPXoAEPcfQ?format=jpg&name=large>|f.jpg
- The process completed apparently successfully but the process as a whole **could not be considered reliable**. Therefore, **the function 'grabimages' was run directly**

as a shell script without being mediated by a Python programme. This time, the process was completed smoothly and reliably

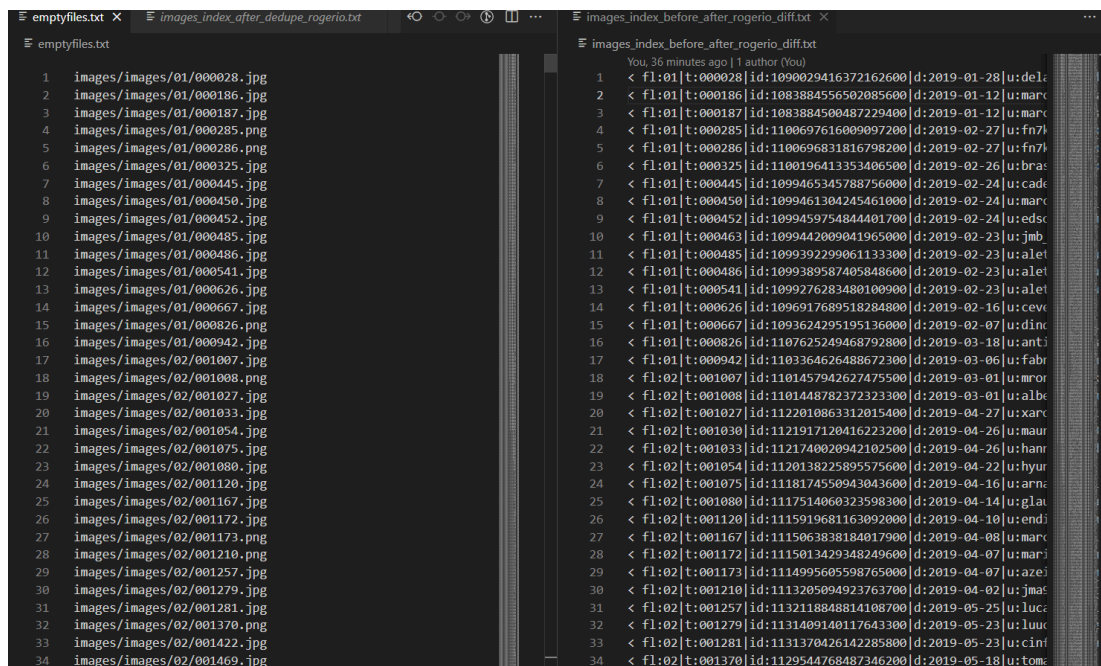
Function 'removedupes'

Determining the empty images

- The empty images were determined by running the following command
 - A list of 1954 empty images was obtained

```
In [ ]: (my_env) ubuntu@ip-172-31-46-151:~/my_env/cl_images$ find images/images -type f
```

- Before running 'removedupes', the file 'images_index.txt' had 40065 lines with the image index matching the number of the corresponding line
- After running 'removedupes':
 - the function did not find duplicated posts because the auxiliary file 'd' was empty
 - all of the empty images were removed
 - the file 'images_index.txt' had 38459 lines (the image index no longer matching the number of the corresponding line). Therefore, $(40065 - 38459 = 1606)$ images have been removed. That means that 12 files other than the empty ones were removed - this is the case of images 000463 and 001030. Both of them have an image link in the field 'f'
 - fl:01|t:000463|id:1099442009041965000|d:2019-02-23|u:jmb_newagel|i:https://pbs.twimg.com/ext_tw_video_thumb/1099395251335
 - fl:02|t:001030|id:1121917120416223200|d:2019-04-26|u:mauroba15398952|i:<https://pbs.twimg.com/media/D5Gg0EmXsAMmdNe.jl>



- The following command was used to create a list of images that 'removedupes' removed, 'images_index_before_after_rogerio_diff.txt'

```
In [ ]: (my_env) ubuntu@ip-172-31-46-151:~/my_env/cl_images_yara$ diff images_index_befo
```

- Comparing with the results obtained by Yara, her 'images_index_after_dedupe_yara.txt' contains 38603. That means that she obtained (38603 - 38459 = 144) more images. One possible explanation is that the Twitter users removed a few posts since Yara collected the images

Function 'uploadtobucket'

- The function completed successfully
- The code required a slight adaptation to run over Ubuntu

```
In [ ]: (my_env) ubuntu@ip-172-31-46-151:~/my_env/cl_images$ more nohup.out
Directory ./images/images uploaded to laelimages/images successfully!
(my_env) ubuntu@ip-172-31-46-151:~/my_env/cl_images$
```

Function 'googlelabels'

Exception handling 1

- An exception was raised while processing the image 008136.jpg, located in the row i=7926:

```
In [ ]: ubuntu@ip-172-31-46-151:~/my_env/cl_images$ tail nohup.out
---- detect-labels 7923 / 40065 ----
```

```

---- detect-labels 7924 / 40065 ----
---- detect-labels 7925 / 40065 ----
---- detect-labels 7926 / 40065 ----
Traceback (most recent call last):
  File "/home/ubuntu/my_env/cl_images/climages.py", line 93, in <module>
    labels = detect_labels(image_uri)
  File "/home/ubuntu/my_env/cl_images/climages.py", line 58, in detect_labels
    raise Exception(f'Error: {response.error.message}')
Exception: Error: Internal server error. Failed to process features.
ubuntu@ip-172-31-46-151:~/my_env/cl_images$

```

- There was nothing wrong with image 008136.jpg
- The function 'googlelabels' was suspended due to an exception reported by the Google Vision service. The service experienced an internal error and failed. It means that the error occurred at Google Vision's side
- In order to resume the processing from the point it stopped, the server was rebooted just to get it freshly started and the function 'googlelabels' was adjusted as follows:

```

In [ ]: for i in range(7926, last + 1):
        try:
            with open('images/images_index.txt', 'r', encoding='utf8') as f:
                line = next(line for j, line in enumerate(f, start = 1) if j == i)
                folder = line.split('|')[0][3:]
                n = line.split('|')[1][2:]
                id = line.split('|')[2][3:]
                file = line.split('|')[5][2:]
                ext = line.split('|')[6][2:5]

            print(f"---- detect-labels {i} / {last} ----")

```

- The processing was resumed successfully

Takeaways

- This exception shed light on the logic of the function 'googlelabels'
 - The counter variable 'i' holds the row of the file 'images_index.txt'. Since the function 'removedupes' removed a few images, the number of the rows no longer matches the index of the images. This explains the discrepancy between the counter 'i' and the name of the image
 - The variable 'last', on the other hand, holds the index of the last row of the file 'images_index.txt'. In our case, it is set to 40065. It means that the processing of labels will try to iterate beyond the quantity of images to be processed because now 'images_index.txt' contains 38459 images. Therefore, when the function 'googlelabels' tries to process the row 38460, it will not find any values and the programme will break
- In spite of this, by the time the programme reaches this breaking point, its purpose will have been served. Therefore, we will manage this as an exception and live with the problem

Exception handling 2

- An exception was raised while processing the image 020425.jpg, located in the row i=19759:

```
In [ ]: ubuntu@ip-172-31-46-151:~/my_env/cl_images$ tail nohup.out
----- detect-labels 19756 / 40065 -----
----- detect-labels 19757 / 40065 -----
----- detect-labels 19758 / 40065 -----
----- detect-labels 19759 / 40065 -----
Traceback (most recent call last):
  File "/home/ubuntu/my_env/cl_images/climages.py", line 93, in <module>
    labels = detect_labels(image_uri)
  File "/home/ubuntu/my_env/cl_images/climages.py", line 58, in detect_labels
    raise Exception(f'Error: {response.error.message}')
Exception: Error: Internal server error. Failed to process features.
ubuntu@ip-172-31-46-151:~/my_env/cl_images$
```

- This was exactly the same error handled in 'Exception handling 1' and was solved with the same solution:

for i in range(19759, last + 1):

Takeaways

- This second occurrence of the same error on Google Vision's side may indicate an intentional behaviour, a mechanism through which Google could save resources by reassessing the customer's intention to continue to process data with their API. Apparently, this presumed mechanism is evoked every ~12 thousand API calls. If that is true, either:
 - the issue could be solved by monitoring the server regularly and take action to resume the processing or;
 - the 'googlelabels' function could be adapted to respond automatically to this error
- The usual rate of Google Vision processing is 4 images processed per minute. However, there is spike in performance when the API calls are resumed - for a few minutes, the API sustains a rate of 30 images per minute. This behaviour could be Google's way of saving resources when its API is called repeatedly

```

ec2-15-229-49-55.sa-east-1.compute.amazonaws.com ***
-rw-rw-r-- 1 ubuntu ubuntu 1674 Nov 23 03:07 020402.txt
-rw-rw-r-- 1 ubuntu ubuntu 651 Nov 23 03:07 020403.txt
-rw-rw-r-- 1 ubuntu ubuntu 2185 Nov 23 03:07 020404.txt
-rw-rw-r-- 1 ubuntu ubuntu 1515 Nov 23 03:07 020405.txt
-rw-rw-r-- 1 ubuntu ubuntu 1524 Nov 23 03:08 020406.txt
-rw-rw-r-- 1 ubuntu ubuntu 2805 Nov 23 03:08 020407.txt
-rw-rw-r-- 1 ubuntu ubuntu 1348 Nov 23 03:08 020408.txt
-rw-rw-r-- 1 ubuntu ubuntu 92 Nov 23 03:08 020409.txt
-rw-rw-r-- 1 ubuntu ubuntu 2654 Nov 23 03:09 020410.txt
-rw-rw-r-- 1 ubuntu ubuntu 2239 Nov 23 03:09 020411.txt
-rw-rw-r-- 1 ubuntu ubuntu 1992 Nov 23 03:09 020412.txt
-rw-rw-r-- 1 ubuntu ubuntu 2789 Nov 23 03:09 020413.txt
-rw-rw-r-- 1 ubuntu ubuntu 836 Nov 23 03:09 020414.txt
-rw-rw-r-- 1 ubuntu ubuntu 1912 Nov 23 03:10 020415.txt
-rw-rw-r-- 1 ubuntu ubuntu 1230 Nov 23 03:10 020416.txt
-rw-rw-r-- 1 ubuntu ubuntu 929 Nov 23 03:10 020417.txt
-rw-rw-r-- 1 ubuntu ubuntu 3266 Nov 23 03:10 020418.txt
-rw-rw-r-- 1 ubuntu ubuntu 1512 Nov 23 03:11 020419.txt
-rw-rw-r-- 1 ubuntu ubuntu 1895 Nov 23 03:11 020420.txt
-rw-rw-r-- 1 ubuntu ubuntu 2444 Nov 23 03:11 020421.txt
-rw-rw-r-- 1 ubuntu ubuntu 1490 Nov 23 03:11 020422.txt
-rw-rw-r-- 1 ubuntu ubuntu 1895 Nov 23 03:12 020423.txt
-rw-rw-r-- 1 ubuntu ubuntu 3192 Nov 23 03:12 020424.txt
-rw-rw-r-- 1 ubuntu ubuntu 1594 Nov 23 13:26 020425.txt
-rw-rw-r-- 1 ubuntu ubuntu 2753 Nov 23 13:26 020426.txt
-rw-rw-r-- 1 ubuntu ubuntu 2506 Nov 23 13:26 020427.txt
-rw-rw-r-- 1 ubuntu ubuntu 1730 Nov 23 13:26 020428.txt
-rw-rw-r-- 1 ubuntu ubuntu 2183 Nov 23 13:26 020429.txt
-rw-rw-r-- 1 ubuntu ubuntu 1022 Nov 23 13:26 020430.txt
-rw-rw-r-- 1 ubuntu ubuntu 2248 Nov 23 13:27 020431.txt
-rw-rw-r-- 1 ubuntu ubuntu 2291 Nov 23 13:27 020432.txt
-rw-rw-r-- 1 ubuntu ubuntu 1507 Nov 23 13:27 020433.txt
-rw-rw-r-- 1 ubuntu ubuntu 1911 Nov 23 13:27 020434.txt
-rw-rw-r-- 1 ubuntu ubuntu 1415 Nov 23 13:27 020435.txt
-rw-rw-r-- 1 ubuntu ubuntu 2741 Nov 23 13:27 020436.txt
-rw-rw-r-- 1 ubuntu ubuntu 1986 Nov 23 13:27 020437.txt
-rw-rw-r-- 1 ubuntu ubuntu 2356 Nov 23 13:27 020438.txt
-rw-rw-r-- 1 ubuntu ubuntu 3192 Nov 23 13:27 020439.txt
-rw-rw-r-- 1 ubuntu ubuntu 4160 Nov 23 13:27 020440.txt
-rw-rw-r-- 1 ubuntu ubuntu 477 Nov 23 13:27 020442.txt
-rw-rw-r-- 1 ubuntu ubuntu 2867 Nov 23 13:27 020443.txt
-rw-rw-r-- 1 ubuntu ubuntu 2364 Nov 23 13:27 020444.txt
-rw-rw-r-- 1 ubuntu ubuntu 1595 Nov 23 13:27 020445.txt
-rw-rw-r-- 1 ubuntu ubuntu 2657 Nov 23 13:27 020446.txt
-rw-rw-r-- 1 ubuntu ubuntu 2190 Nov 23 13:27 020447.txt
-rw-rw-r-- 1 ubuntu ubuntu 1803 Nov 23 13:27 020448.txt
-rw-rw-r-- 1 ubuntu ubuntu 753 Nov 23 13:27 020449.txt
-rw-rw-r-- 1 ubuntu ubuntu 2491 Nov 23 13:27 020450.txt
-rw-rw-r-- 1 ubuntu ubuntu 2373 Nov 23 13:27 020451.txt
-rw-rw-r-- 1 ubuntu ubuntu 2721 Nov 23 13:27 020452.txt
-rw-rw-r-- 1 ubuntu ubuntu 568 Nov 23 13:27 020453.txt
-rw-rw-r-- 1 ubuntu ubuntu 2352 Nov 23 13:27 020454.txt
-rw-rw-r-- 1 ubuntu ubuntu 753 Nov 23 13:27 020455.txt
-rw-rw-r-- 1 ubuntu ubuntu 383 Nov 23 13:27 020456.txt
-rw-rw-r-- 1 ubuntu ubuntu 2576 Nov 23 13:27 020457.txt
-rw-rw-r-- 1 ubuntu ubuntu 753 Nov 23 13:27 020458.txt
-rw-rw-r-- 1 ubuntu ubuntu 3047 Nov 23 13:27 020459.txt
-rw-rw-r-- 1 ubuntu ubuntu 2002 Nov 23 13:27 020460.txt
-rw-rw-r-- 1 ubuntu ubuntu 753 Nov 23 13:27 020461.txt
-rw-rw-r-- 1 ubuntu ubuntu 2093 Nov 23 13:28 020462.txt
-rw-rw-r-- 1 ubuntu ubuntu 2168 Nov 23 13:28 020463.txt
-rw-rw-r-- 1 ubuntu ubuntu 1415 Nov 23 13:28 020464.txt
-rw-rw-r-- 1 ubuntu ubuntu 2286 Nov 23 13:28 020465.txt
-rw-rw-r-- 1 ubuntu ubuntu 3411 Nov 23 13:28 020467.txt
-rw-rw-r-- 1 ubuntu ubuntu 768 Nov 23 13:28 020468.txt
-rw-rw-r-- 1 ubuntu ubuntu 2678 Nov 23 13:28 020469.txt
-rw-rw-r-- 1 ubuntu ubuntu 2743 Nov 23 13:28 020470.txt

```

Exception handling 3

An exception was raised while processing the image 035822.jpg, located in the row i=34470:

```

In [ ]: ubuntu@ip-172-31-46-151:~/my_env/cl_images$ tail nohup.out
---- detect-labels 34467 / 40065 ----
---- detect-labels 34468 / 40065 ----
---- detect-labels 34469 / 40065 ----
---- detect-labels 34470 / 40065 ----
Traceback (most recent call last):
File "/home/ubuntu/my_env/cl_images/climages.py", line 93, in <module>
    labels = detect_labels(image_uri)

```

```
File "/home/ubuntu/my_env/cl_images/climages.py", line 58, in detect_labels
    raise Exception(f'Error: {response.error.message}')
Exception: Error: Internal server error. Failed to process features.
ubuntu@ip-172-31-46-151:~/my_env/cl_images$
```

- This was exactly the same error handled in 'Exception handling 1' and 'Exception handling 2' and was solved with the same solution:

```
for i in range(34470, last + 1):
```

Takeaways

This was the third occurrence of the same error. It happened after ~15 thousand API calls. Definitely, this is a standard behaviour of Google Vision.

Exception handling 4

In the function 'datamatrix' the command 'awk' is used to create the document 'data.csv'. When executing, 'awk' exceeds the limit of memory use and is killed by the operating system. During the troubleshooting, it has been determined that this issue was being caused by the particularly large amount of data to process.

```
In [ ]: --- v000999 ---
        --- v001000 ---
        --- images/data.csv .....
datamatrix.sh: line 31: 5823 Killed                  awk '
    FNR==1 { col++ }
    FNR>max { max=FNR }
    { l[FNR,col]=$0 }
    END {
        for (i=1;i<=max;i++) {
            for (j=1;j<=col;j++) {
                printf "%-50s",l[i,j]
            }
            print ""
        }
    }
' images/temp/* > u
ubuntu@ip-172-31-46-151:~/my_env/cl_images$
```

A portion of 1000 images was processed in order to allow the inspection of the file 'data.csv'. It has been determined that it was created by the juxtaposition of a series of input files as follows:

Therefore, a valid solution would be to develop a Python programme to juxtapose the files.

The new solution via 'pandas' produced exactly the same file 'data.csv':

When running with Yara's data the same memory overflow occurred. At the end of the day, the cause of the problem was insufficient system memory. The EC2 instance type that was being used, 't2.micro', was equipped with just 1 GiB of memory. When the code was transferred to a larger EC2 instance type, 'r5.large', with 16 GiB of memory, the function 'datamatrix' completed successfully.

Lessons learned

Yara's project was the first opportunity to employ Jupyter Notebooks to process Corpus Linguistics research data. Here are a few reflections on the lived experience

1. The issues faced by Yara were probably due to the limited performance of a personal computer and home broadband access in the face of heavy workloads. When bottlenecks hit a personal computer, files may end up corrupted
2. While Jupyter Notebooks are highly didactic, its processing relies on a Python kernel whose performance cannot be compared to a Python interpreter running over a Linux server. Jupyter Notebooks are ideal for testing ideas and prototyping solutions, they may not be so when it comes to dealing with heavy workloads
3. Cloud Computing is probably a better solution for GELC's research projects due to following reasons:
 - The projects pay only for the resources the utilise. There is no need to source and maintain hardware
 - The computing service is provided by server-grade virtual machines that are dedicated and tuned to the required workload
 - It provides easy-to-deploy infrastructure to process Big Data workloads
4. Appropriate server sizing is crucial, especially considering that some functions are memory intensive

The following figure shows the monitoring of the server of this project when it was running the 'googl labels' function. The CPU utilisation remained steady at 10%.

